MIL-C-60559D(MU)
8 July 1970
SUPERSEDING
MIL-C-60559C(MU)
15 April 1969

MILITARY SPECIFICATION
CARTRIDGE, IMPULSE: M225

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers Cartridge, Impulse: M225, for use in conjunction with Launcher and Grenade Smoke, H.C. and W.P., XM176.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification For
MIL-C-60539	- Coating Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-810	- Environmental Test Methods

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command

D10535474	- Cartridge, Impulse, M225 Assembly
C9206812	- Carton, Packing, Ammunition for Cartridge, Impulse, M225
C9206813	- Box, Packing, Ammunition for Cartridge, Impulse, M225
IEL-10535474	- Inspection Equipment List for Cartridge, Impulse, M225

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASA B46.1 - Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-STD-810, Figure 514-5, without evidence of any disintegration of the primer, propellant or derangement of components.

3.3 Primer sensitivity.- The M29A1 Primer shall be used and shall function under an applied energy of 15.7 inch ounces, maximum and shall not fire under an applied energy of 1.9 inch ounces, minimum.

3.4 Charge weight.- The amount of propellant charge by weight in any one cartridge shall not be in excess or less that two percent of the established charge.

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3.5 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.6 Functional.

3.6.1 The cartridge shall not fail to fire.

3.6.2 Ballistics.- The cartridge when temperature conditioned over a temperature range of -20° to -25° Fahrenheit (F), 65° to 75° and 125° to 130°F and fired in M176 launcher containing two inert grenades H.C. and W.P. shall comply with the requirements prescribed in Table I. Grenade launcher angle shall be 34 plus or minus 2 degrees with the horizontal. Grenade launcher heights shall be 77 plus or minus 1 inch from the muzzle to the horizontal plane.

TABLE I

Temperature Degree Fahrenheit (F)	Range per feet Min.	Max.	Ignition Delay Milliseconds Max. (See 6.2)
-20° to -25°F	100	150	10
65° to 75°F	100	150	10
125° to 130°F	100	150	10

3.7 Sealant application.- The application of sealant in any one cartridge shall be in accordance with Drawing D10535474.

3.8 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.8.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.8.2 Burr.- The part or assembly shall not have a burr which might interfere with assembly or function of the item or which might be injurious to personnel handling the item.

3.8.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services comform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 50 cartridge cases without protective finish and 50 seal closures without protective finish for hardness tests, 23 cartridge cases and 23 seal closures for protective finish tests, 60 cartridges for ballistic tests, 100 primed cartridges which shall be tested as indicated in Table II herein and 10 sets of unassembled components (including explosive components) for visual and dimensional examination.

Table II

Paragraphs Rqmts	Test	Test	Sample Size	Type of Test Dest.	Non-Dest.
Dwg.	4.4.1	Airtightness	60 (from ballistics)	-	X
3.2	4.4.2	Vibration	12 (from ballistics)	-	X
3.3	4.4.3	Primer Sensitivity	100 cartridges	X	-
3.4	4.4.4	Charge Weight	50 (from no-fire primer sensitivity test)	X	-
3.5	4.4.5	Protective finish	23 cartridge cases	X	-
			23 closure seals	X	-
Dwg.	4.4.6	Hardness	50 cartridge cases (without protective finish)	X	-
			50 closure seals (without protective finish)	X	-
3.6.2	4.4.7.1	Ballistics	60 (as shown in Table III)	X	-
3.7	4.4.8	Sealant Application	50 (from no-fire primer sensitivity test)	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

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4.2.1.1 Examination and tests.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE III

Temperature	Not Vibrated	Vibrated
-20° to -25°F	16	4
65° to 75°F	16	4
125° to 130°F	16	4

4.2.1.3 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- Unless otherwise specified by the contracting officer, formation of lots, lot size, and presentation of lots, shall be in accordance with the "Submission of Product" provisions of MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, an ammunition lot shall consist of an identifiable group of cartridges consisting of:

a. Components produced by a homogeneous process
b. Propellant from not more than one lot
c. Primers from not more than one lot

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor group, the criteria in Table V shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

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4.3.2 (cont'd)

TABLE IV

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 301	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE V

Total Allowable Defects Per Sample Size

Sample Size for Major Defects	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
over - 43	6	8	11	14
Sample Size for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
over - 48	8	11	17	20

4.3.2.1 Primer, M29A1, Drawing C8798312, covering a detail of Drawing D10535474.

Defect	Method of Inspection
CRITICAL	
1. Mixed primers	Visual
2. Primer pellet broken, cracked or disloged	Visual
3. Primer foil discolored	Visual
4. Metal defective	Visual

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4.3.2.1 (Cont'd)

Defects	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.3.2.2 Case, Drawing C10535472, covering a detail of Drawing D10535474.

CRITICAL	
1. Depth of primer pocket	Gage
2. Metal defective	Visual
3. Primer flash hole missing or obstructed	Visual
MAJOR	
101. Concentricity of head with primer pocket	Gage
102. Concentricity of body with primer pocket	Gage
103. Diameter of head, min.	Gage
104. Diameter of primer pocket	Gage
105. Diameter of flash hole	Gage
106. Diameter of body, min.	Gage
107. Thickness of head	Gage
108. Protective finish inadequate or defective	Visual
109. Surface finish of primer pocket improper	Visual
110. Burr	Visual
111. Foreign matter	Visual
MINOR	
201. Diameter of head, max.	Gage
202. Diameter of body, max.	Gage
203. Diameter of cavity	Gage
204. Depth of cannelure	Gage
205. Total length	Gage
206. Thickness of web	Gage
207. Location of cannelure	Gage
208. Radius or chamfer missing or incorrect	Visual
209. Surface finish improper, except as otherwise classified	Visual

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4.3.2.3 Closure Seal, Drawing B10535473, covering a detail of Drawing D10534747.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Diameter, min.	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max.	Gage
202. Total length	Gage
203. Thickness of sidewall radius	Gage
204. Thickness of ruptured area	Gage
205. Radius missing or incorrect	Visual
206. Surface finish improper	Visual

4.3.2.4 Cartridge, Impulse, M225, Assembly, Drawing D10535474.

CRITICAL	
1. Primer not properly seated	Gage
2. Any part missing	X-ray and Visual 1/
3. Improper assembly	X-ray and Visual 1/
4. Mixed ammunition	Visual
5. Crimp missing or incorrect	Visual
6. Marking missing, incorrect or illegible	Visual
7. Metal defective	Visual
8. Closure sealant missing or improperly applied	Visual
MAJOR	
101. Diameter of body, min.	Gage
102. Diameter of head, min.	Gage
103. Length of head flange	Gage
104. Protective finish inadequate or defective	Visual
105. Primer seal missing or improperly applied	Visual
106. Burr	Visual
107. Foreign matter	Visual

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4.3.2.4 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Diameter of body, max.	Gage
202. Diameter of head, max.	Gage
203. Total length	Gage
204. Fold, wrinkle, buckle, bulge, dent or draw scratch	Visual
205. Varnish missing or inadequate	Visual

1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negative, properly identified, shall be forwarded to the procuring activity.

4.3.2.5 Packing in carton, Drawing C9206812.

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Contents loose	Manual & Visual
202. Cut or damaged through all carton layers	Visual
203. Marking missing, incorrect or illegible	Visual
204. Tape incomplete or badly wrinkled	Visual
205. Color of tape incorrect	Visual

4.3.2.6 Packing in Wooden Box, Drawing C9206813.

CRITICAL

None defined

MAJOR

None defined

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4.3.2.6 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Contents loose	Manual and Visual
202. Hardware missing, loose, broken or improperly engaged	Manual and Visual
203. Strapping missing, loose, broken or improperly assembled	Manual and Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4:

4.3.3.1 Airtightness.- Sixty cartridges of each production lot shall be subjected to this test. Failure of any cartridge to comply with the requirements of the drawing shall be cause for rejection of the lot of cartridges.

4.3.3.2 Vibration.- Twelve of the 60 assembled cartridges of the initial production lot shall be subjected to the vibration test. After vibration, the 12 cartridges shall be x-rayed in accordance with Note 1/ 4.3.2.4. If any change is made in the lot of primers or propellants in any subsequent production lot, an additional sample of 12 cartridges shall be selected and vibrated as above. Failure of any cartridge to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.3 Primer sensitivity.- One hundred primed cartridges from the initial production lot shall be tested, using 50 cartridges for the all-fire test and 50 cartridges for the No-fire test. Further sensitivity tests shall be performed when there is a change in the cartridge case or primer lot. Failure of any primer not to fire in the all-fire test or any primer to fire in the no-fire test, shall be cause for rejection of the lot of cartridges in accordance with 3.3.

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4.3.3.4 Charge weight.- The 50 cartridges from the no-fire primer sensitivity test (4,4.3) shall be used for this test. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.5 Protective finish.- Twenty three units of each anodized component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.5 shall be cause for rejection of the lot.

4.3.3.6 Hardness.- Fifty cartirdge cases and fifty closure seals from each heat-treated lot shall be tested. Failure of any unit to comply with the requirements of the drawing shall be cause for rejection of the lot.

4.3.3.7 Functional.- The following tests shall be performed by the Government:

4.3.3.7.1 Ballistics.- Sixty cartridges from each lot shall be grouped and temperature conditioned as specified in Table VI and 4.4.7.1.1 and then tested in accordance with 4.4.7.1. Failure of any cartridge to comply with the requirement of 3.6.1 and 3.6.2 shall be cause for rejection of the lot.

TABLE VI

Temperature	Sample Size
-20° to -25°F	20
65° to 75°F	20
125° to 130°F	20

4.3.3.8 Sealant application.- The 50 cartridges from the no-fire primer sensitivity test (4.4.3) shall be used for this test. Failure of any cartridge to comply with the requirements of 3.7 shall be cause for rejection of the lot.

4.3.4 Inspection equipment.- The examination and tests shall be performed using the equipment listed on IEL-10535474.

4.4 Test methods and procedures.

4.4.1 Airtightness.- The method and procedure for determining airtightness shall be by use of equipment listed on the applicable inspection equipment list.

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4.4.2 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-STD-810, Figure 514-5 modified as follows: 5 to 14 cycles per second (cps) at 0.4 inch double amplitude, 14 to 500 cps at 4 acceleration of gravity (gs). The cycling method shall be 5 cps to 500 cps to 5 cps in 15 minutes for a total of:

a. One hundred minutes at 65° to 75°F in each of 2 directions.

b. Fifty minutes at -20° to -25°F in each of 2 directions.

c. Fifty minutes at 125° to 130°F in each of 2 directions.

4.4.3 Primer sensitivity.- Remove the propellant charge from the test cartridges by sectioning the cartridge around the periphery approximately .625 inches from head end. The primer sensitivity test shall be made using an all-fire and no-fire test. The all-fire drop height shall be 8.01 inches and the no-fire drop height shall be .99 inches. One half of the prescribed sample of 100 cartridges shall be tested, using a 15/16 inch diameter ball weighing 1.94 ± .002 ounces at each of the two heights, using a primer sensitivity machine equipped with a cartridge holder approved by the procuring activity. The head space shall be zero. All height measurements shall be made between the head of the firing pin and the bottom of the suspended ball. The primer sensitivity machine shall be checked to be sure the drop ball is impacting the center of the firing pin. All firing pin indent centers shall be within 0.02 inches of the center of the primer.

4.4.4 Charge weight.- The propellant charge from each sectioned cartridge of 4.4.3 shall be placed in individual containers, examined, weighed and the results recorded.

4.4.5 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539. The cartridge case and seal closure shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.6 Hardness.- Prior to the application of protective finish, a hardness testing machine shall be used to determine compliance with the hardness requirements specified on the applicable drawing.

4.4.7 Functional.

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4.4.7.1 Ballistics.- The cartridge sample shall be tested in M176 launchers within 5 minutes after removal from the conditioning chamber. When this time is exceeded, the sample shall be reconditioned in accordance with 4.4.7.1.1.

The M176 launcher shall be mounted at an angle of 34 plus or minus 2 degrees. The luancher muzzle elevation shall be 77 inches plus or minus 1 inch above horizontal. The distance from the luancher base end to the initial grenade impact shall be recorded for each grenade in accordance with 3.6.2.

4.4.7.1.1 Temperature conditioning.- A dummy M225 cartridge containing a temperature sensitive element, shall be placed in the conditioning chamber at the same time as the samples. The conditioning time shall be one hour longer than the time necessary to bring the element contained in the dummy cartrdige to the required temperature.

4.4.8 Sealant application.- The sectioned cartridges (of 4.4.3) shall be examined to determine that the propellant cavity is free of sealant.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C9206812 and C9206813.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the title, number and date of this specification.

6.2 Ignition delay is defined as the time that elapsed between the firing of the primer and grenade rupture of launcher closure disc.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A400

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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